A right rectangular prism is 6 cm by 4 cm by 5 cm.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q1/5

A right prism has a base area of 12 square feet and a height of 3 feet.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q2/5

A right triangular prism has a base area of 10 square inches and a length of 8 inches.

What is its volume?

A. 40 cubic inches

B. 80 cubic inches

C. 18 cubic inches

D. 800 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q3/5

True or false?

A cube with edges of 10 m has a volume of 1,000 cubic meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q4/5

True or false?

A 2 by 5 by 10 box has a volume of 17 cubic units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q5/5

SET 52 • 8.G.C.9

FORWARD 1

Model real objects with combinations of solids • High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1

A capsule-shaped vitamin is a cylinder 1 cm long with a radius of 0.3 cm, with half of a sphere of radius 0.3 cm on each end.

What solids make up the model?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q1

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q1/5

A pencil is modeled as a cylinder with a cone at one end.

Which measurements are needed to find its volume?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q2

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q2/5

Which solids best model a traffic cone sitting on a square base?

A. A cone on a square prism

B. A sphere on a cylinder

C. Two cylinders

D. A cube on a pyramid

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q3

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q3/5

True or false?

A model can use simpler shapes even when the real object is not exactly that shape.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q4

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q4/5

True or false?

A water bottle is best modeled as one sphere.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q5

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q5/5

SET 52 • 8.G.C.9

FORWARD 2

Apply volume to design and capacity problems • High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

A company wants a cylinder-shaped can that holds 500π cubic cm with a radius of 5 cm.

How tall must the can be?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q1

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q1/5

A cone-shaped funnel has a radius of 6 cm and a height of 10 cm. Water drains at 6π cubic cm per second.

How long does a full funnel take to drain?

Time: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q2

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q2/5

Each can is a cylinder holding the same volume.

Which design uses the most metal for the curved side?

A. Radius 4 cm, height 6.25 cm

B. Radius 2.5 cm, height 16 cm

C. Radius 5 cm, height 4 cm

D. Radius 2 cm, height 25 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q3

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q3/5

True or false?

Two cans with the same volume can have different surface areas.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q4

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q4/5

True or false?

A taller can always holds more than a shorter can.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q5

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q5/5

SET 53 • 8.SP.A.1 • STATISTICS & PROBABILITY

Construct a scatter plot for bivariate measurement data

MAIN	Grade 8 main question	**8.SP.A.1**
BACKWARD 1	Graph points that represent a real-world situation	**5.G.A.2**
BACKWARD 2	Display a numerical data set in a dot plot	**6.SP.B.4**
FORWARD 1	Represent two-variable data on a scatter plot and describe how t…	**HSS-ID.B.6**
FORWARD 2	Fit an exponential or quadratic function to data and use it to make	**HSS-ID.B.6.a**

SET 53 • 8.SP.A.1

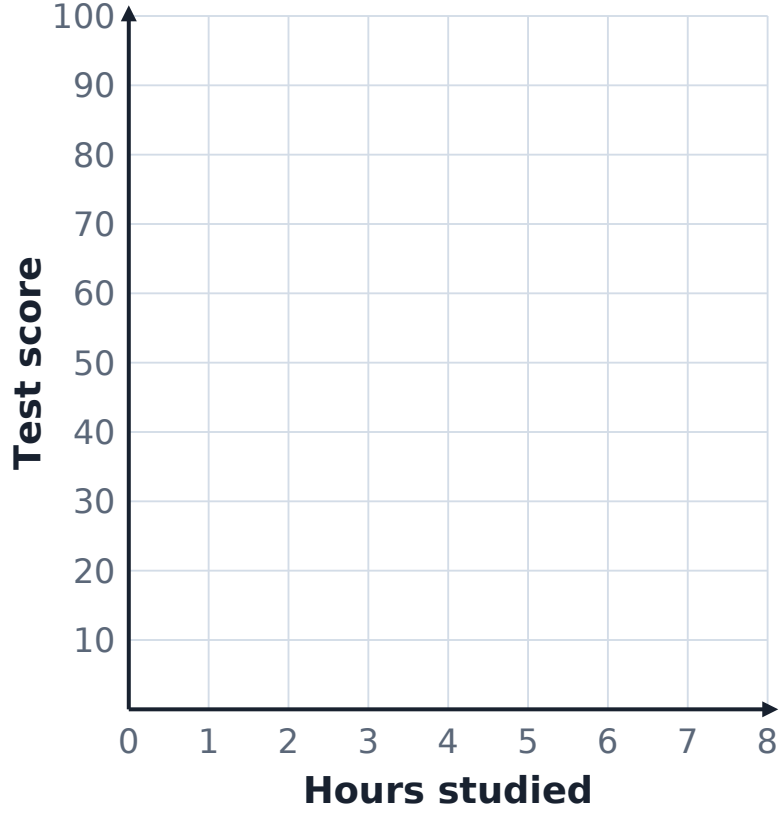
MAIN

Grade 8 main question • Grade 8 • Statistics & Probability • 8.SP.A.1

Seven students recorded (hours studied, test score):

(1, 50), (2, 60), (3, 60), (4, 70), (5, 80), (6, 80), (7, 90)

Make a scatter plot of the data on the grid.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

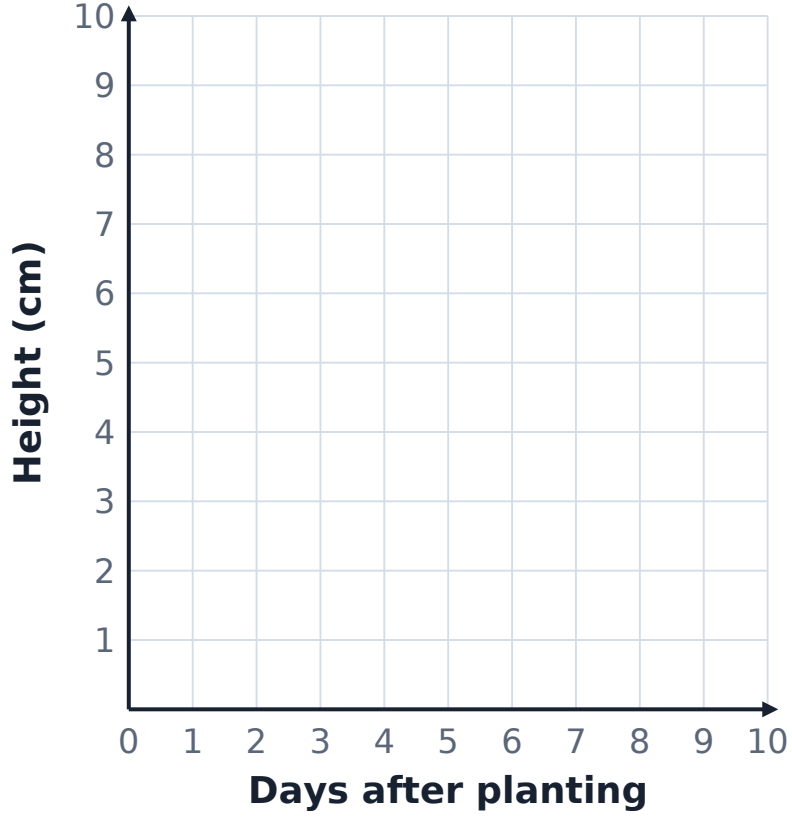
ID S53-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q1/5

A bean plant was measured on different days (days after planting, height in cm):

(2, 1), (4, 3), (5, 4), (6, 4), (8, 6), (9, 7), (10, 8)

Make a scatter plot of the data on the grid.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

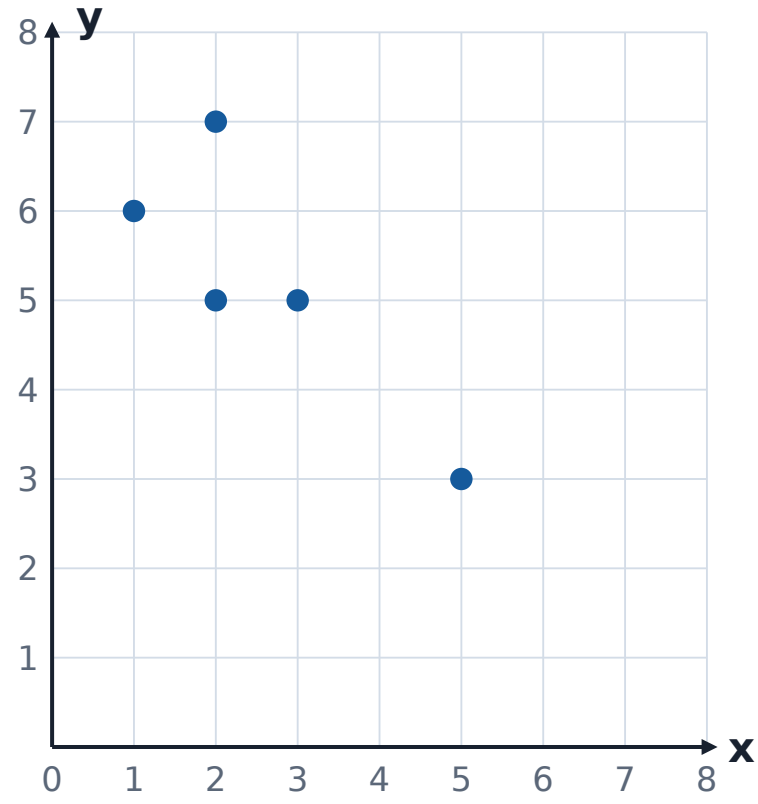
ID S53-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q2/5

The scatter plot should show the data (1, 6), (2, 5), (3, 5), (5, 3), and (7, 2), but one point is plotted wrong.

Which point is wrong, and where should it be?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q3/5

Ten students recorded their shoe sizes and heights to make a scatter plot.

What does each point on the scatter plot show?

A. How many students have each shoe size

B. One student's shoe size and height

C. The mean shoe size of the class

D. One student's shoe size only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q4/5

True or false?

A scatter plot of hours slept and test scores puts two points on the grid for each student, one for each variable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q5/5

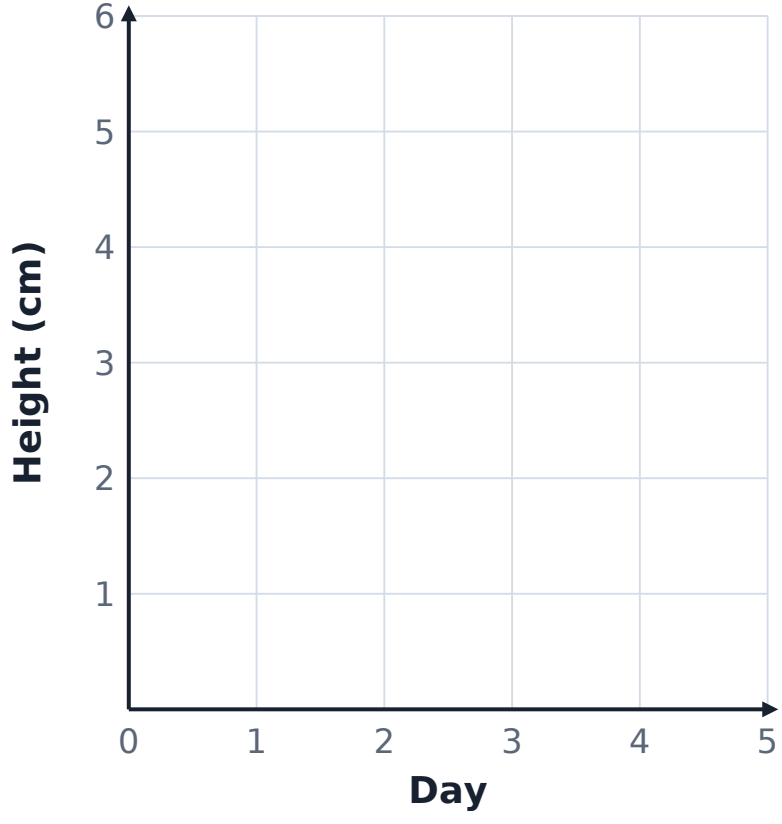
SET 53 • 8.SP.A.1

BACKWARD 1

Graph points that represent a real-world situation • Grade 5 • Geometry • 5.G.A.2

A plant was 2 cm tall on day 1, 4 cm tall on day 2, and 5 cm tall on day 3.

Plot the points (day, height) on the grid.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q1/5

A graph shows points (day, height in cm).

What does the point (4, 3) mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q2/5

Points on a graph are (hours, miles).

Which point shows 5 hours and 2 miles?

A. (5, 5)

B. (2, 5)

C. (5, 2)

D. (2, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q3/5

True or false?

To plot (3, 6), start at the origin, move 3 units right, then 6 units up.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q4/5

True or false?

The point (0, 4) is on the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q5/5

SET 53 • 8.SP.A.1

BACKWARD 2

Display a numerical data set in a dot plot • Grade 6 • Statistics & Probability • 6.SP.B.4

The numbers of pets of 10 students are 0, 1, 1, 2, 2, 2, 3, 3, 4, and 1.

Make a dot plot of the data on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q1/5

How many students are shown in the dot plot?



Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q2/5

Which display shows each value of a data set as a dot above a number line?

A. A box plot

B. A bar graph

C. A histogram

D. A dot plot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q3/5

True or false?

In a dot plot, a value that appears 3 times has 3 dots stacked above it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q4/5

True or false?

A dot plot shows two different measurements for each student.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q5/5

SET 53 • 8.SP.A.1

FORWARD 1

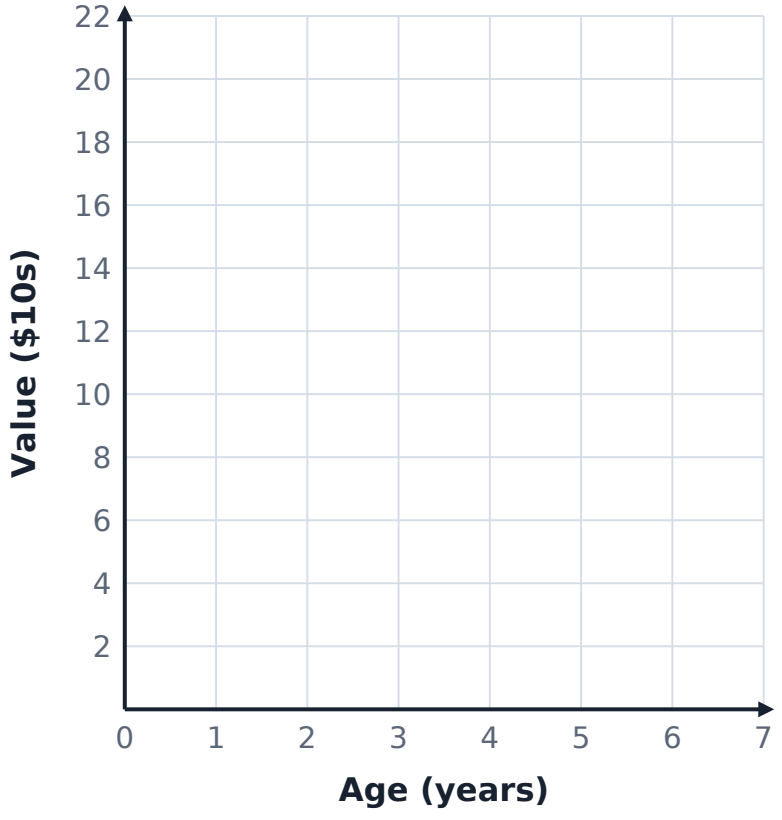
Represent two-variable data on a scatter plot and describe how the variables are related • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6

Six used bikes have these (age in years, value in tens of dollars):

(1, 20), (2, 17), (3, 15), (4, 12), (5, 10), (6, 8)

Make a scatter plot of the data. Then describe how age and value are related.

Relationship: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q1/5

A scatter plot of hours of sleep and reaction time shows points falling from left to right.

Describe how the variables are related.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q2/5

Which pair of variables would most likely show a positive association on a scatter plot?

A. Height and arm span of adults

B. Outside temperature and heating bill

C. Shoe size and test score

D. Age of a car and its value

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F1-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q3/5

True or false?

A scatter plot whose points rise from left to right shows a positive association.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q4/5

True or false?

If two variables have no association, their scatter plot points lie on a horizontal line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q5/5

SET 53 • 8.SP.A.1

FORWARD 2

Fit an exponential or quadratic function to data and use it to make predictions • High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

A bacteria count is 100 at hour 0, 200 at hour 1, 400 at hour 2, and 800 at hour 3.

Write a function that fits the data.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q1/5

A bacteria count fits y = 100 · 2^x after x hours.

Predict the count at hour 6.

Count: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q2/5

Which function fits the data exactly?

x	0	1	2	3	4
y	0	1	4	9	16

A. $y = x^2$

B. $y = 4x$

C. $y = x + 4$

D. $y = 2^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q3

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q3/5

True or false?

The data (0, 5), (1, 10), (2, 20), and (3, 40) fit $y = 5 \cdot 2^x$ exactly.

A. True

B. False

True or false?

Data whose y-values go up by the same amount at each step are best fit by an exponential function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q5/5

SET 54 • 8.SP.A.1 • STATISTICS & PROBABILITY

Describe clustering, outliers, and positive, negative, linear, or nonlinear association in a scatter plot

MAIN	Grade 8 main question	**8.SP.A.1**
BACKWARD 1	Describe a data distribution by its center, spread, and shape	**6.SP.A.2**
BACKWARD 2	Describe the overall pattern of data and any striking deviations	**6.SP.B.5.c**
FORWARD 1	Distinguish between correlation and causation	**HSS-ID.C.9**
FORWARD 2	Interpret the correlation coefficient of a linear fit	**HSS-ID.C.8**

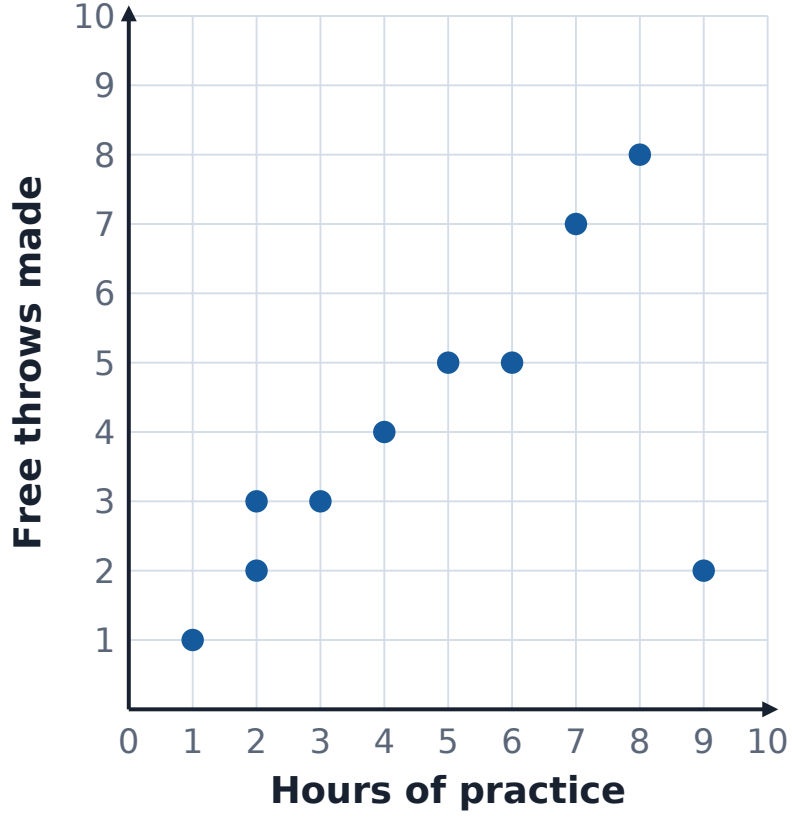
SET 54 • 8.SP.A.1

MAIN

Grade 8 main question • Grade 8 • Statistics & Probability • 8.SP.A.1

Describe the association in the scatter plot (positive, negative, or none; linear or nonlinear), and name any outlier.

Answer: _____



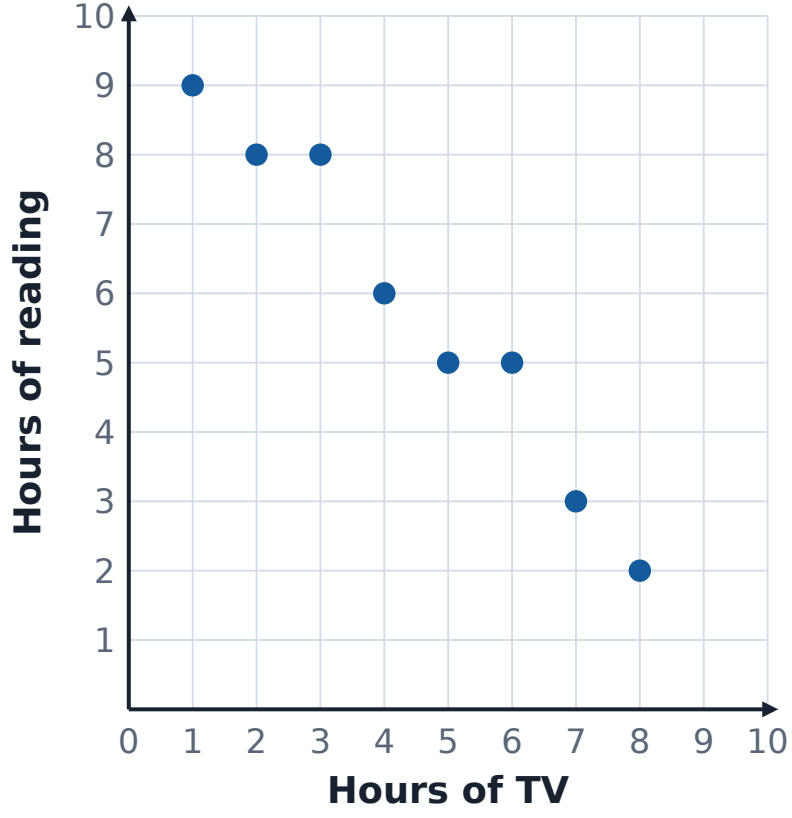
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q1/5

Describe the association in the scatter plot.

Answer: _____

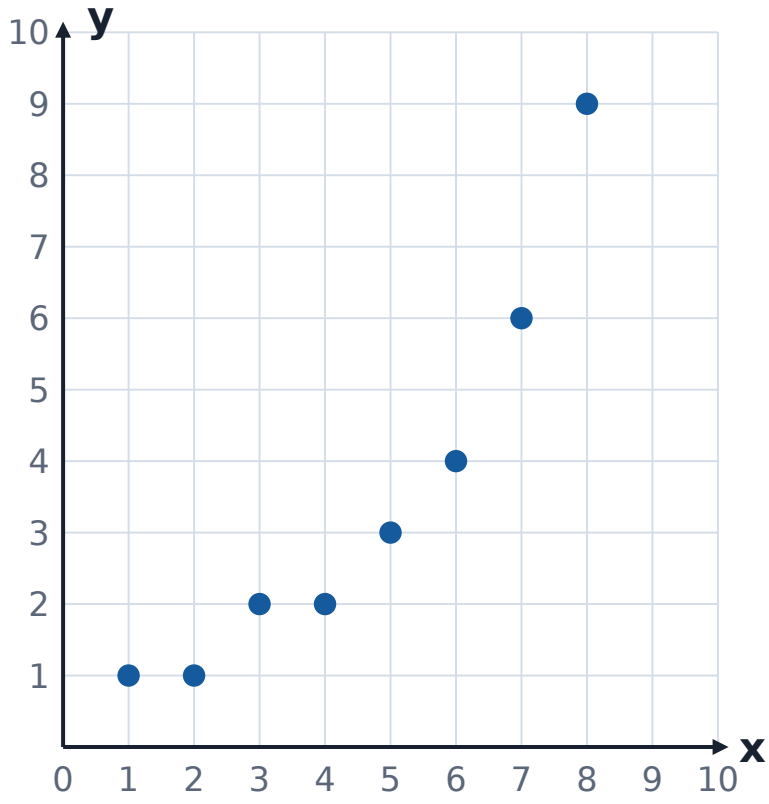


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q2/5

Which best describes the association in the scatter plot?



A. Positive and linear

B. No association

C. Negative and nonlinear

D. Positive and nonlinear

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

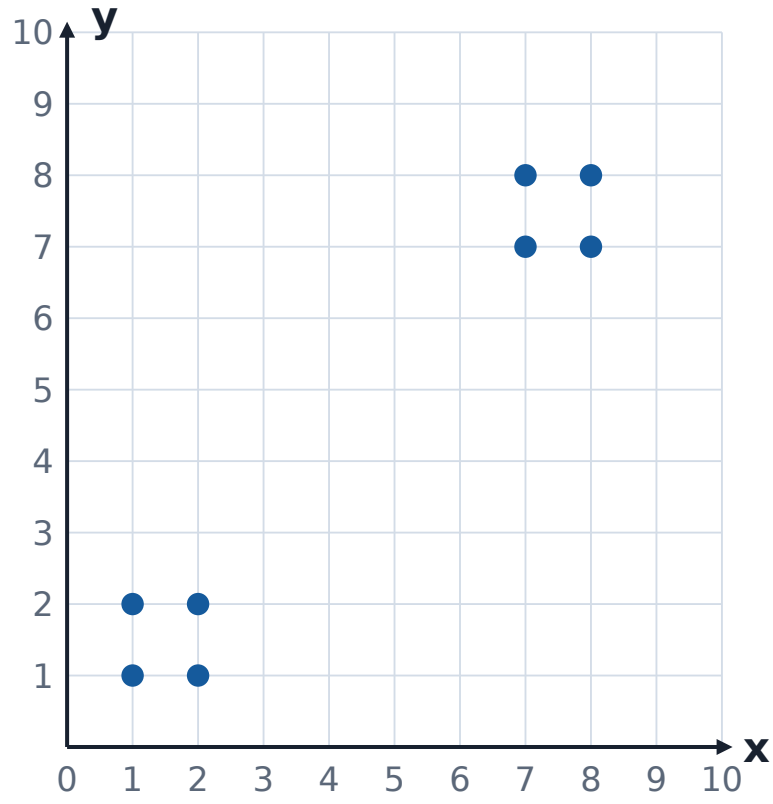
ID S54-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q3/5

The points in the scatter plot form two groups.

Describe the two clusters.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q4

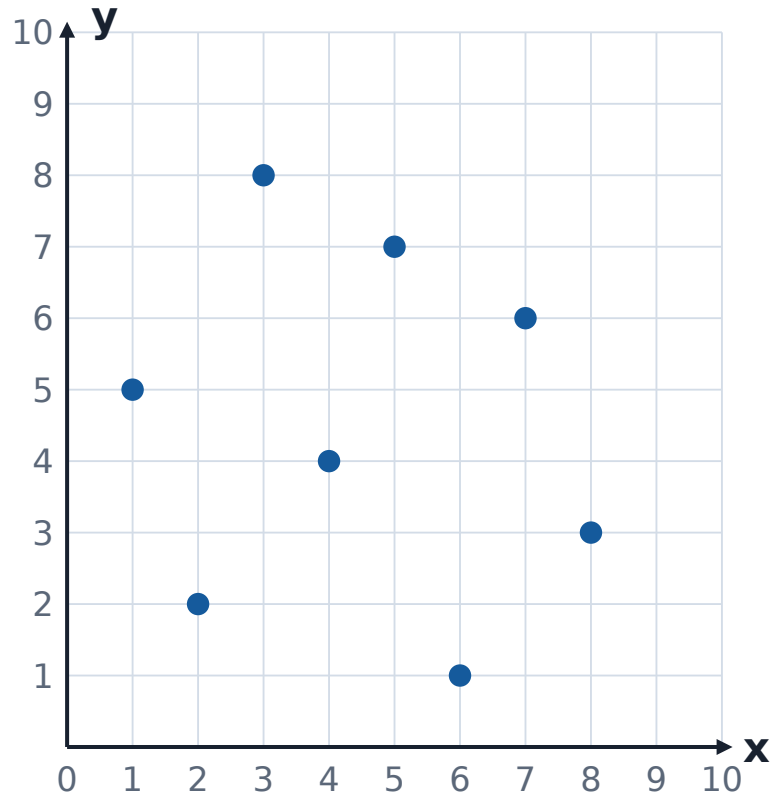
Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q4/5

True or false?

The scatter plot shows a negative association.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q5

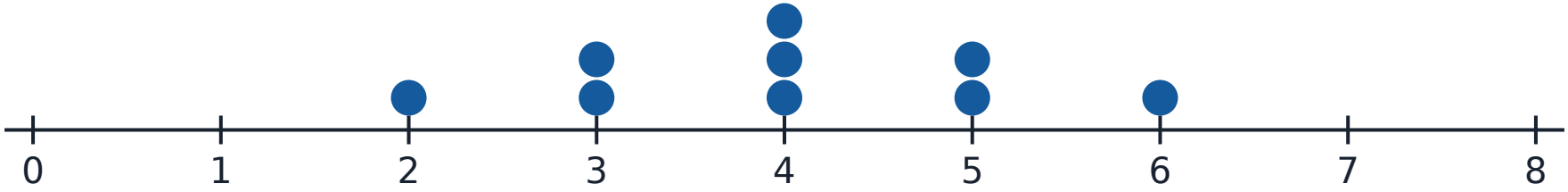
Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q5/5

SET 54 • 8.SP.A.1

BACKWARD 1

Describe a data distribution by its center, spread, and shape • Grade 6 • Statistics & Probability • 6.SP.A.2

Is the data in the dot plot symmetric or skewed?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q1/5

What is the range of the data in the dot plot?



Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q2/5

Which value is far from the rest of the data?

3, 4, 4, 5, 5, 6, 18

A. 18

B. 6

C. 3

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q3/5

True or false?

The data set 2, 3, 3, 4, 4, 4, 5, 5, 6 is symmetric about 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q4/5

True or false?

The range of a data set measures its center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q5/5

SET 54 • 8.SP.A.1

BACKWARD 2

Describe the overall pattern of data and any striking deviations • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the median.

4, 6, 7, 7, 8, 9, 30

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q1/5

Which value is a striking deviation from the pattern?

4, 6, 7, 7, 8, 9, 30

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q2/5

The data are 10, 12, 12, 13, 14, and 40.

Which measure of center is changed more by the value 40?

A. The mean

B. The median

C. Both are changed equally

D. Neither is changed

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q3/5

True or false?

Removing a value that is much larger than the others lowers the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q4/5

True or false?

An outlier always changes the median by a large amount.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q5/5

SET 54 • 8.SP.A.1

FORWARD 1

Distinguish between correlation and causation • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

Cities with more ice-cream sales also have more sunburns.

Does buying ice cream cause sunburns? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q1/5

A study finds that students who sleep more tend to have higher grades.

Can you conclude that more sleep causes higher grades? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q2/5

Which kind of study can show that one variable causes a change in another?

A. An observational study

B. A randomized experiment

C. A survey of volunteers

D. A scatter plot of survey data

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F1-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q3/5

True or false?

Two variables can be strongly associated without one causing the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q4/5

True or false?

A strong correlation shows that one variable causes the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q5/5

SET 54 • 8.SP.A.1

FORWARD 2

Interpret the correlation coefficient of a linear fit • High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

The correlation coefficient of a data set is $r = -0.92$.

Describe the association.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q1/5

Which shows the stronger linear association, $r = 0.45$ or $r = -0.81$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q2/5

Points lie almost exactly on a line that rises from left to right.

Which value of r fits?

A. -0.98

B. 0.98

C. 0

D. 0.10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q3**

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q3/5

True or false?

The correlation coefficient r is always between −1 and 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q4/5

True or false?

$r = -0.9$ shows a weaker linear association than $r = 0.5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q5/5

SET 55 • 8.SP.A.2 • STATISTICS & PROBABILITY

Informally fit a straight line to a scatter plot and judge how closely it fits

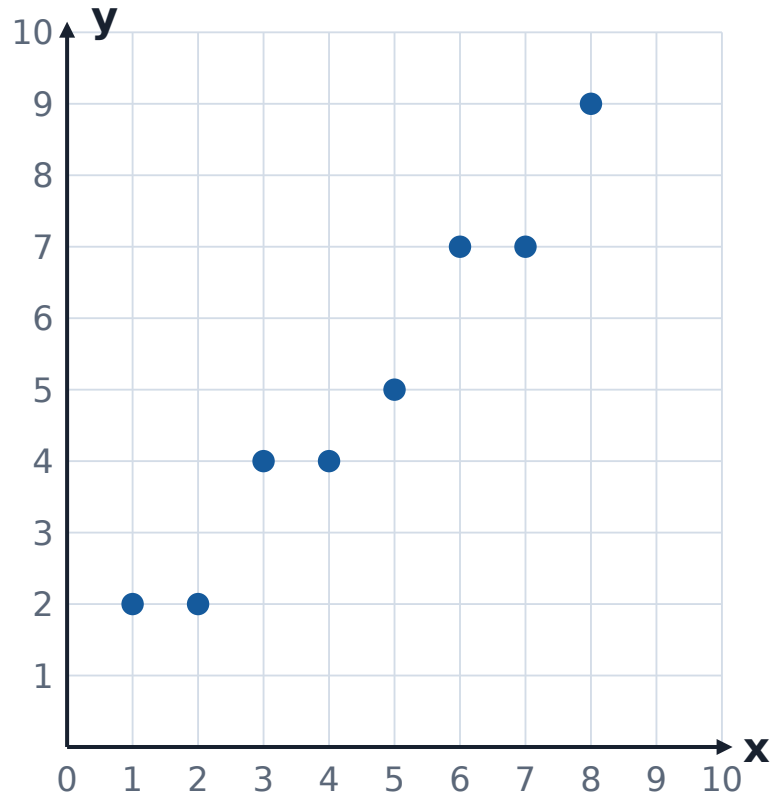
MAIN	Grade 8 main question	**8.SP.A.2**
BACKWARD 1	Decide from a table or graph whether a relationship is proportion…	**7.RP.A.2.a**
BACKWARD 2	Write an equation for a relationship shown in a table or graph	**6.EE.C.9**
FORWARD 1	Fit a linear function to data that suggest a linear association	**HSS-ID.B.6.c**
FORWARD 2	Judge the fit of a line by finding and plotting residuals	**HSS-ID.B.6.b**

SET 55 • 8.SP.A.2

MAIN

Grade 8 main question • Grade 8 • Statistics & Probability • 8.SP.A.2

Draw a straight line that fits the data. Then write an equation of your line.



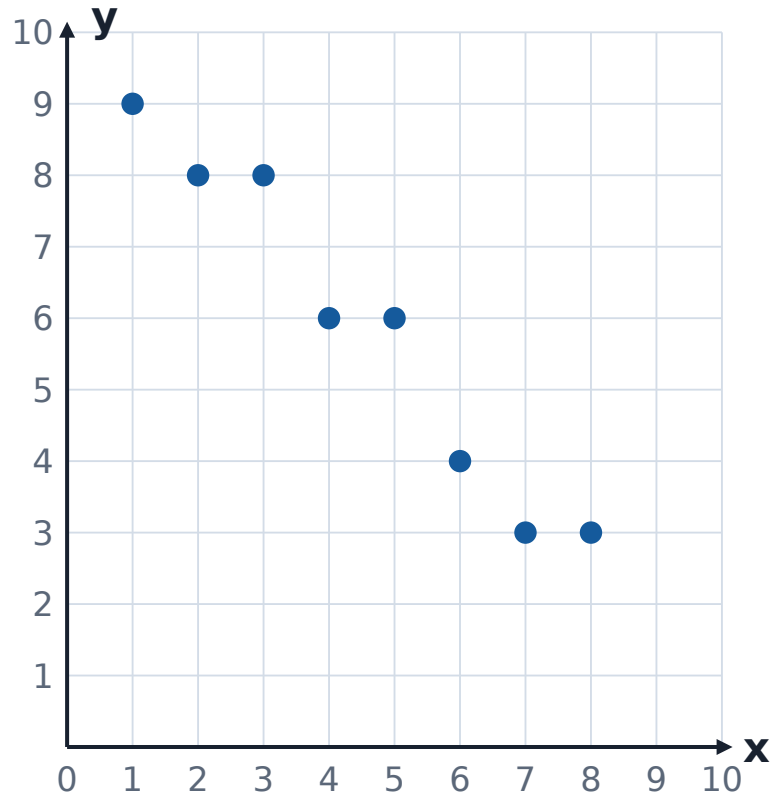
Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q1/5

Draw a straight line that fits the data. Use your line to estimate y when x = 9.



Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q2/5

Two students drew lines for the same scatter plot. Line A has every point above it. Line B has about half the points above it and half below, all close to it.

Which line fits better? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q3/5

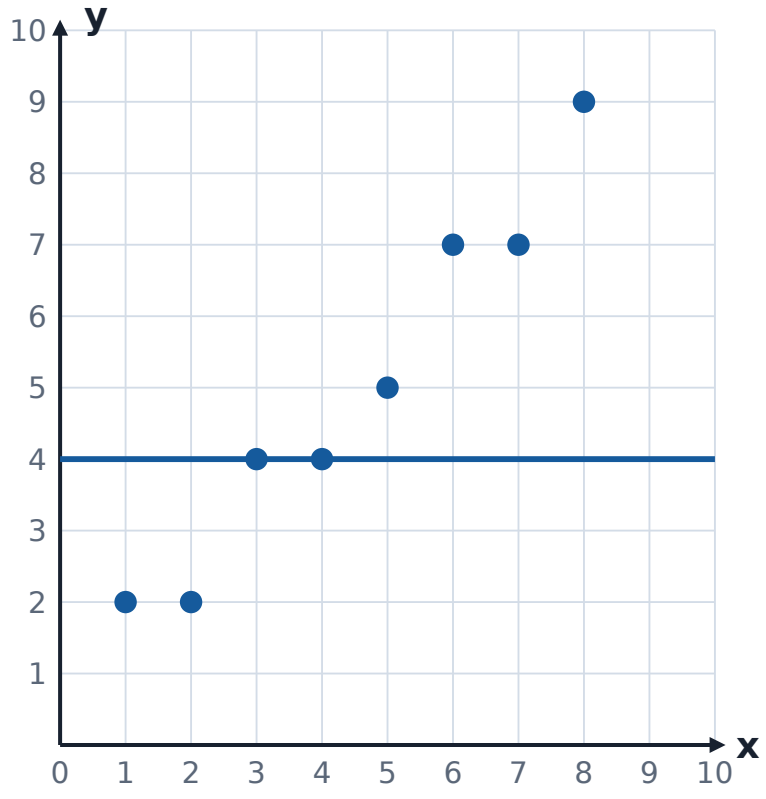
Why is the line on the scatter plot a poor fit?

A. It is not long enough.

B. It does not follow the upward trend of the points.

C. It starts at the y-axis.

D. It passes through too many points.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q4/5

True or false?

A line that fits a scatter plot well must pass through every point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q5/5

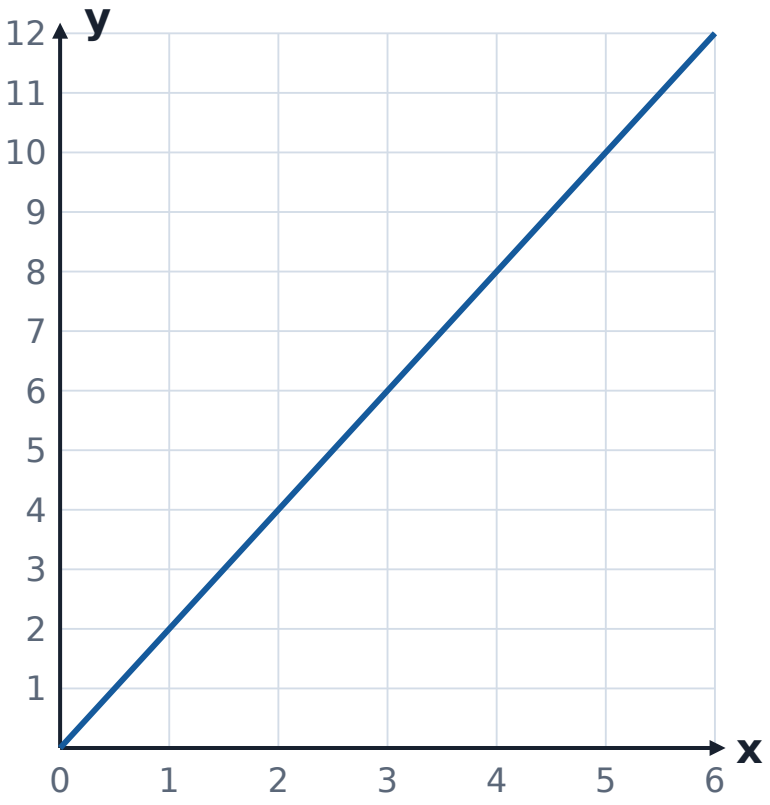
SET 55 • 8.SP.A.2

BACKWARD 1

Decide from a table or graph whether a relationship is proportional • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

Is the relationship shown in the graph proportional? Explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q1/5

Is the relationship in the table proportional? Explain.

x	1	2	4
y	3	6	13

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q2/5

Which graph shows a proportional relationship?

A. A curve through (0, 0) and (2, 4)

B. A straight line through (0, 2) and (2, 6)

C. A horizontal line through (0, 3)

D. A straight line through (0, 0) and (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q3/5

True or false?

A straight line through (0, 0) and (3, 9) shows a proportional relationship.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q4/5

True or false?

Every straight-line graph shows a proportional relationship.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q5/5

SET 55 • 8.SP.A.2

BACKWARD 2

Write an equation for a relationship shown in a table or graph • Grade 6 • Expressions & Equations • 6.EE.C.9

Write an equation for y in terms of x.

x	1	2	3
y	4	8	12

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q1/5

Write an equation for y in terms of x.

x	0	1	2	3
y	2	3	4	5

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q2/5

A line passes through (0, 0), (1, 5), and (2, 10).

Which equation does it show?

A. y = 5x

B. y = 10x

C. x = 5y

D. y = x + 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q3/5

True or false?

The points (1, 3), (2, 6), and (3, 9) fit y = 3x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q4/5

True or false?

The points (0, 1), (1, 2), and (2, 3) fit y = 2x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q5/5

SET 55 • 8.SP.A.2

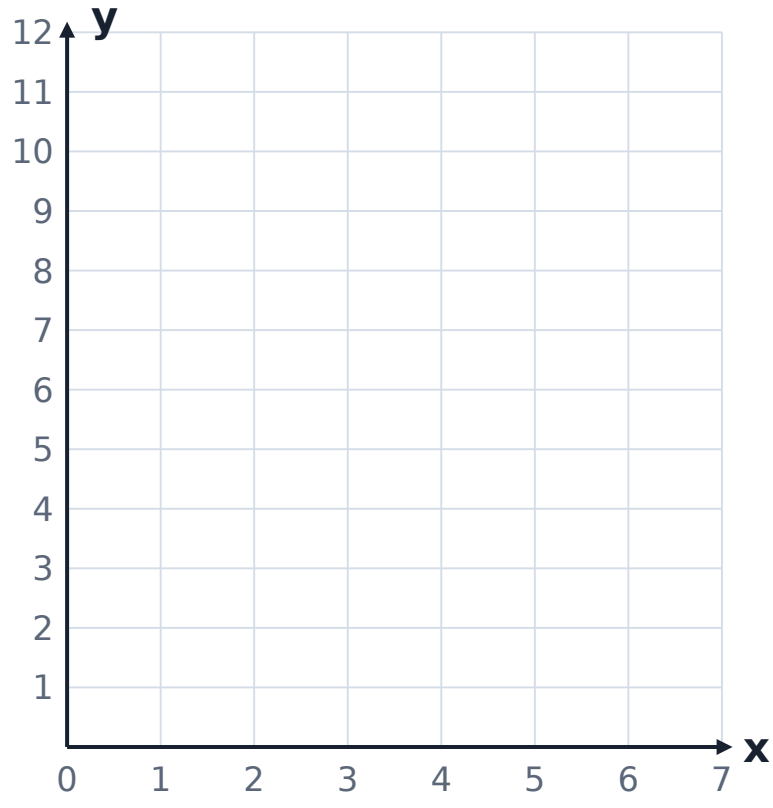
FORWARD 1

Fit a linear function to data that suggest a linear association • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c

The data are (1, 3), (2, 5), (3, 6), (4, 8), (5, 10), and (6, 11).

Make a scatter plot, draw a line that fits the data, and write an equation of your line.

Equation: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q1/5

A line of fit passes through (2, 5) and (6, 13).

Write its equation.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q2/5

A line of fit for (hours, dollars) data is $y = 12x + 8$.

What does it predict for 5 hours?

A. $100

B. $68

C. $25

D. $60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F1-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q3/5

True or false?

A line of fit through (0, 4) and (4, 12) has a slope of 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q4/5

True or false?

A line of fit should always be drawn through the first and last data points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q5/5

SET 55 • 8.SP.A.2

FORWARD 2

Judge the fit of a line by finding and plotting residuals • High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b

A line of fit is $y = 2x + 1$. A data point is $(3, 8)$.

Find the residual (actual y minus predicted y).

Residual: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q1/5

A line of fit is $y = 2x + 1$. A data point is $(5, 9)$.

Find the residual (actual y minus predicted y).

Residual: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q2/5

A residual plot shows points scattered above and below 0 with no pattern.

What does this suggest?

A. A curve must be used.

B. Every prediction is exact.

C. The line is a poor fit.

D. The line is a good fit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F2-Q3**

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q3/5

True or false?

A residual of 0 means the data point lies on the line of fit.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q4/5

True or false?

A residual plot with a clear U shape shows that a linear model fits well.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q5/5

SET 56 • 8.SP.A.3 • STATISTICS & PROBABILITY

Use the equation of a linear model to interpret the slope and intercept and to make predictions

MAIN	Grade 8 main question	**8.SP.A.3**
BACKWARD 1	Explain what a point on the graph of a proportional relationship m…	**7.RP.A.2.d**
BACKWARD 2	Evaluate a formula for given values of its variable	**6.EE.A.2.c**
FORWARD 1	Interpret the slope and intercept of a linear model in context	**HSS-ID.C.7**
FORWARD 2	Evaluate reports and predictions based on data	**HSS-IC.B.6**

SET 56 • 8.SP.A.3

MAIN

Grade 8 main question • Grade 8 • Statistics & Probability • 8.SP.A.3

A line of fit for the height h (inches) of a plant after w weeks is h = 1.5w + 4.

What do the slope and the intercept mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q1/5

A line of fit for the gas g (gallons) left in a car's tank after m miles is $g = 14 - 0.04m$.

Predict the gas left after 200 miles.

Gas left: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q2/5

A model for the water level w (feet) of a pond t days after the rain stops is $w = 12 - 0.3t$.

What does -0.3 mean?

A. The level rises about 0.3 foot each day.

B. The pond is empty after 0.3 day.

C. The level starts at 0.3 foot.

D. The level drops about 0.3 foot each day.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q3/5

A model for the score s after h hours of practice is $s = 6h + 58$.

Predict the score after 5 hours of practice.

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q4/5

True or false?

In the model $y = 3.2x + 15$ for the cost of a taxi ride of x miles, 15 is the cost per mile.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q5/5

SET 56 • 8.SP.A.3

BACKWARD 1

Explain what a point on the graph of a proportional relationship means • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d

The graph of cost (dollars) against pounds is proportional and passes through (1, 3).

What does (1, 3) mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q1/5

A proportional graph of miles against hours passes through (0, 0).

What does (0, 0) mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q2/5

A proportional graph of (gallons, cost) passes through (1, 4).

What is the unit rate?

A. $5 per gallon

B. $4 per gallon

C. $0.25 per gallon

D. $1 per gallon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q3/5

True or false?

On a proportional graph of (hours, pages), the point (1, 20) means 20 pages are read per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q4/5

True or false?

On a proportional graph of (tickets, cost), the point (2, 18) shows the unit rate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q5/5

SET 56 • 8.SP.A.3

BACKWARD 2

Evaluate a formula for given values of its variable • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate 1.5w + 4 when w = 6.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q1/5

Evaluate $300 - 40t$ when $t = 3$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q2/5

Evaluate 6h + 58 when h = 2.

A. 62

B. 128

C. 66

D. 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q3/5

True or false?

When $x = 4$, $3x + 5 = 35$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q4/5

True or false?

When $t = 2$, $10 - 3t = 14$.

A. True

B. False

SET 56 • 8.SP.A.3

FORWARD 1

Interpret the slope and intercept of a linear model in context • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7

A model for the number n of ice pops sold on a day with a high temperature of t °F is $n = 4t - 200$.

What does the slope mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q1/5

A model for ice-pop sales is $n = 4t - 200$, where t is the temperature in °F.

Does the intercept -200 make sense in this situation? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q2/5

A model for a phone's battery level b (percent) after h hours of use is $b = 100 - 12.5h$.

What does 100 mean?

A. The battery lasts 100 hours.

B. The battery charges 100% each hour.

C. The battery loses 100% each hour.

D. The battery starts at 100%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F1-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q3/5

True or false?

In the model $y = 0.8x + 2$, y increases by about 0.8 for each increase of 1 in x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q4/5

True or false?

In a linear model, the intercept always has a sensible meaning in the situation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q5/5

SET 56 • 8.SP.A.3

FORWARD 2

Evaluate reports and predictions based on data • High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A model of children's heights for ages 2 to 10 is $h = 2.5a + 31$ inches at age a.

A report uses it to predict the height of a 40-year-old. Is that reasonable? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q1/5

A line of fit was made from data collected between 60 °F and 90 °F.

A report uses it to predict sales at 70 °F. Is that reasonable? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q2/5

A report says: "People who own more books live longer, so buying books makes you live longer."

What is the main flaw?

A. It counts too few books.

B. It reports a positive association.

C. It treats an association as a cause.

D. It uses a line of fit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q3

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q3/5

True or false?

Predictions far outside the range of the data are less reliable than predictions inside it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q4/5

True or false?

A survey of volunteers always represents the whole population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q5/5

SET 57 • 8.SP.A.4 • STATISTICS & PROBABILITY

Construct a two-way table summarizing data on two categorical variables

MAIN	Grade 8 main question	**8.SP.A.4**
BACKWARD 1	Report the number of observations in a data set	**6.SP.B.5.a**
BACKWARD 2	Solve "how many more" and "how many in all" problems with a sc…	**3.MD.B.3**
FORWARD 1	Find joint, marginal, and conditional relative frequencies in a two…	**HSS-ID.B.5**
FORWARD 2	Use a two-way table to find conditional probabilities and decide…	**HSS-CP.A.4**

SET 57 • 8.SP.A.4

MAIN

Grade 8 main question • Grade 8 • Statistics & Probability • 8.SP.A.4

60 students were asked whether they have a curfew and whether they have chores. 36 have a curfew. 24 of the students with a curfew have chores. 8 of the students without a curfew have chores.

Complete the two-way table.

	Chores	No chores	Total
Curfew			
No curfew			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q1/5

50 students were asked whether they prefer summer or winter. 22 are in Grade 7 and 28 are in Grade 8. 15 of the Grade 7 students and 18 of the Grade 8 students prefer summer.

Complete the two-way table.

	Summer	Winter	Total
Grade 7			
Grade 8			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q2/5

90 people were asked whether they exercise daily and whether they eat breakfast. 40 exercise daily. 30 of the people who exercise daily eat breakfast. 50 of the 90 people eat breakfast.

Complete the two-way table.

	Breakfast	No breakfast	Total
Exercises			
Does not			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q3/5

Ten students answered two questions: "Do you have a pet?" and "Do you have a sibling?" Their answers (pet, sibling) are:

yes–yes, yes–no, no–no, yes–yes, no–yes, yes–yes, no–no, no–yes, yes–no, no–no

Complete the two-way table.

	Sibling	No sibling	Total
Pet			
No pet			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q4/5

True or false?

A completed two-way table for 60 students can have row totals of 36 and 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q5/5

SET 57 • 8.SP.A.4

BACKWARD 1

Report the number of observations in a data set • Grade 6 • Statistics & Probability • 6.SP.B.5.a

In a survey, 12 students chose soccer, 9 chose basketball, and 7 chose tennis.

How many students were surveyed?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q1/5

How many observations are shown in the dot plot?



Observations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q2/5

Each of 25 students named a favorite fruit. 5 different fruits were named.

How many observations are in the data set?

A. 5

B. 125

C. 25

D. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q3/5

True or false?

If 14 students answer yes and 11 answer no, there are 25 observations.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q4/5

True or false?

The data set 3, 3, 5, 7 has 3 observations.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q5

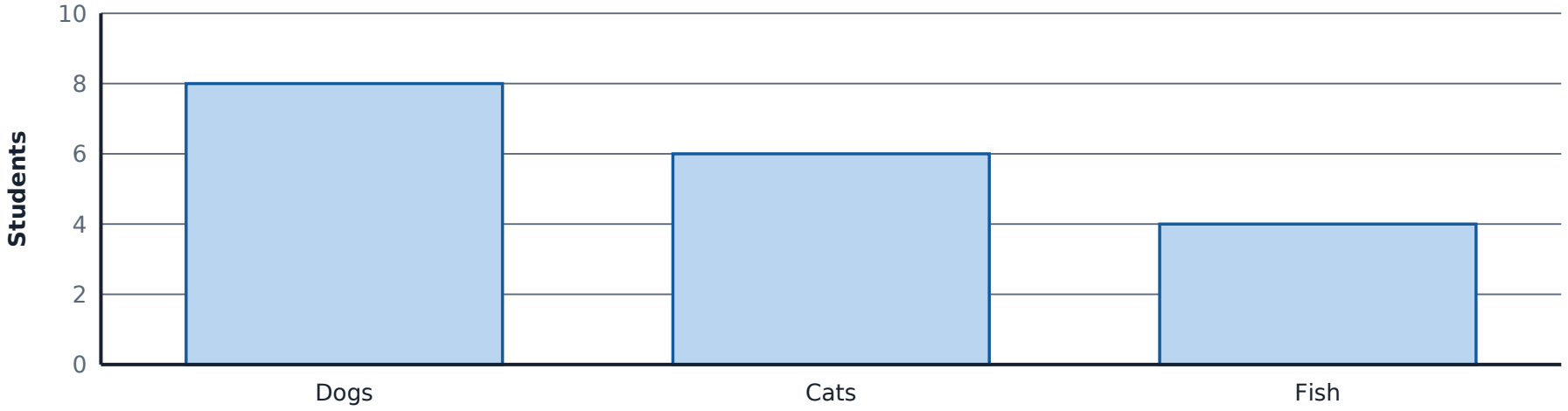
Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q5/5

SET 57 • 8.SP.A.4

BACKWARD 2

Solve "how many more" and "how many in all" problems with a scaled bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

How many more students chose dogs than fish?



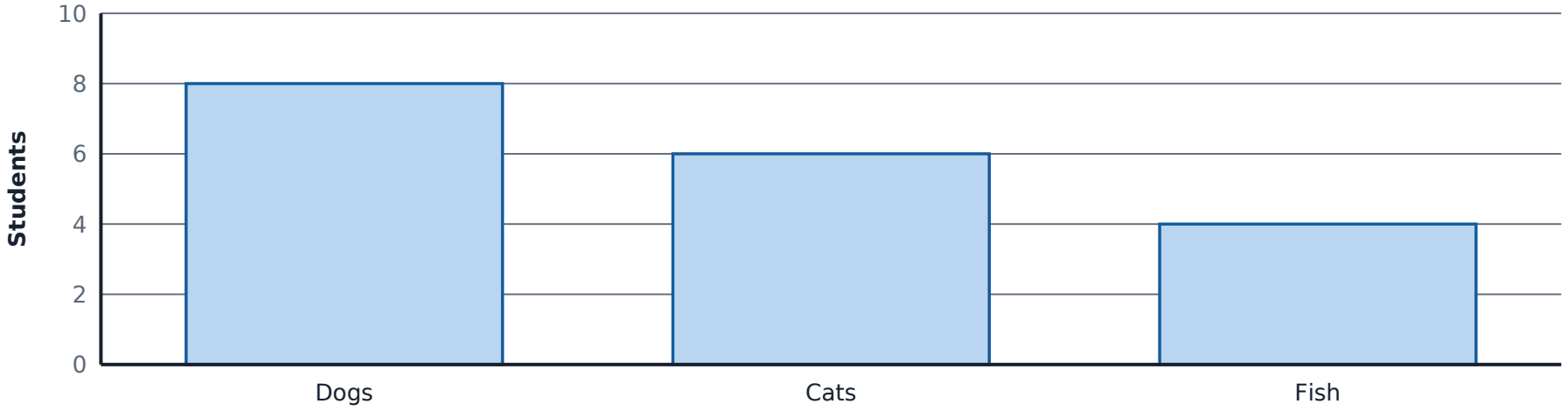
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q1/5

How many students chose a pet in all?



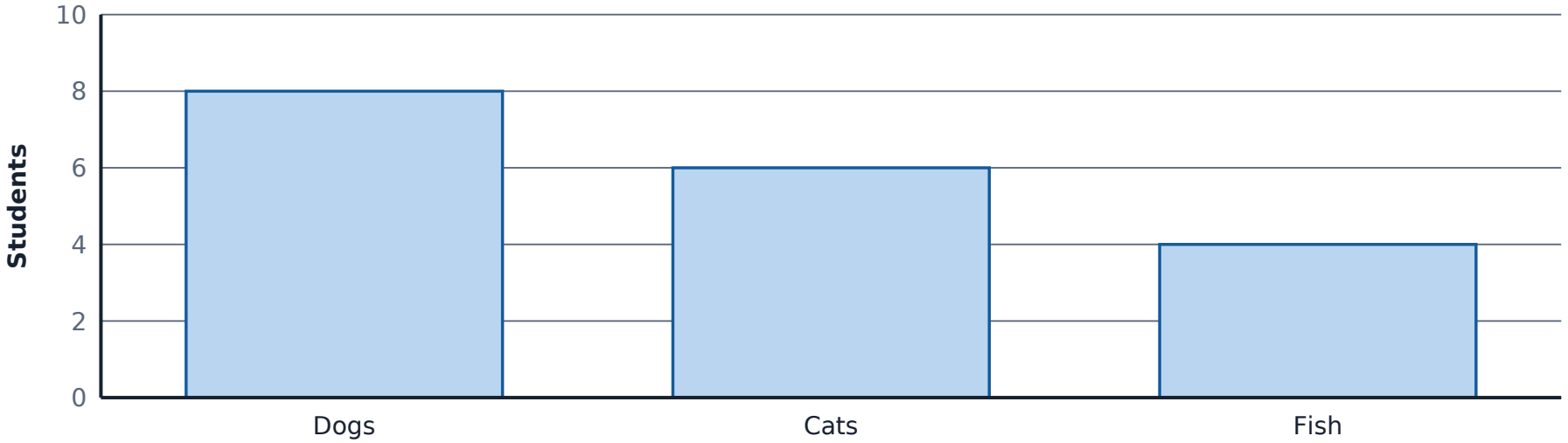
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q2/5

How many fewer students chose fish than cats?



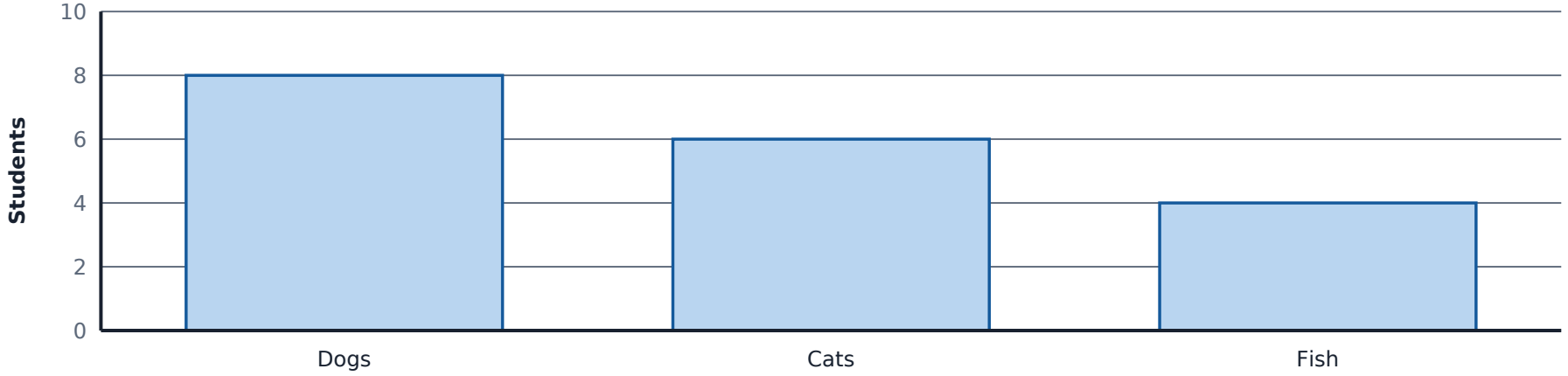
A. 4 **B.** 6 **C.** 10 **D.** 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-B2-Q3**

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q3/5

True or false?

Fewer students chose cats than fish.



A. True

B. False

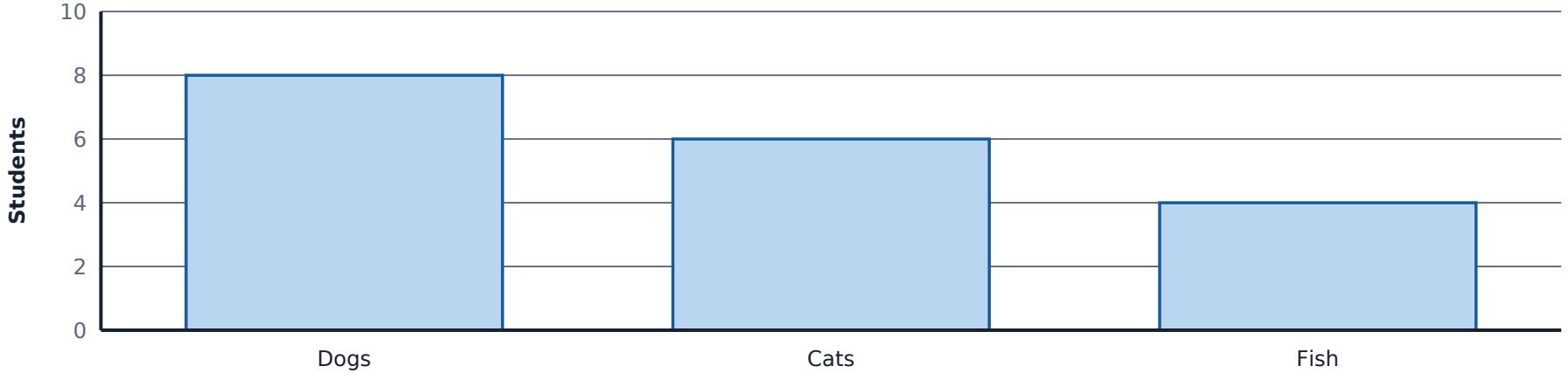
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q4/5

True or false?

The same number of students chose dogs and cats.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q5/5

SET 57 • 8.SP.A.4

FORWARD 1

Find joint, marginal, and conditional relative frequencies in a two-way table • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

What is the joint relative frequency of students who have a curfew and have chores?

	Chores	No chores	Total
Curfew	24	12	36
No curfew	8	16	24
Total	32	28	60

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q1/5

What is the marginal relative frequency of students who have chores? Round to the nearest hundredth.

	Chores	No chores	Total
Curfew	24	12	36
No curfew	8	16	24
Total	32	28	60

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q2/5

What is the conditional relative frequency of having chores among students with no curfew?

	Chores	No chores	Total
Curfew	24	12	36
No curfew	8	16	24
Total	32	28	60

A. $\frac{2}{3}$ **B.** $\frac{1}{4}$ **C.** $\frac{1}{3}$ **D.** $\frac{2}{15}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q3/5

True or false?

The marginal relative frequency of students with a curfew is 0.36.

	Chores	No chores	Total
Curfew	24	12	36
No curfew	8	16	24
Total	32	28	60

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q4/5

True or false?

The joint relative frequencies in the four inner cells add to 2.

	Chores	No chores	Total
Curfew	24	12	36
No curfew	8	16	24
Total	32	28	60

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q5/5

SET 57 • 8.SP.A.4

FORWARD 2

Use a two-way table to find conditional probabilities and decide whether events are independent • High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

A student from the table is chosen at random.

Find P(in club | Grade 8).

	In club	Not in club	Total
Grade 7	12	28	40
Grade 8	18	42	60
Total	30	70	100

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q1/5

Are being in the club and being in Grade 7 independent? Explain with conditional probabilities.

	In club	Not in club	Total
Grade 7	12	28	40
Grade 8	18	42	60
Total	30	70	100

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q2/5

A student from the table is chosen at random.

Find P(Grade 7 | in club).

	In club	Not in club	Total
Grade 7	12	28	40
Grade 8	18	42	60
Total	30	70	100

A. 0.6 **B.** 0.12 **C.** 0.4 **D.** 0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-F2-Q3**

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q3/5

True or false?

For a student chosen at random from the table, P(not in club) = 0.7.

	In club	Not in club	Total
Grade 7	12	28	40
Grade 8	18	42	60
Total	30	70	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q4/5

True or false?

For the table, P(in club | Grade 7) equals P(Grade 7 | in club).

	In club	Not in club	Total
Grade 7	12	28	40
Grade 8	18	42	60
Total	30	70	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-F2-Q5**

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q5/5

SET 58 • 8.SP.A.4 • STATISTICS & PROBABILITY

Use relative frequencies in a two-way table to describe a possible association

MAIN	Grade 8 main question	**8.SP.A.4**
BACKWARD 1	Find a percent of a quantity and the percent one number is of ano…	**6.RP.A.3.c**
BACKWARD 2	Use ratio language to describe two quantities	**6.RP.A.1**
FORWARD 1	Recognize conditional probability and independence in everyday…	**HSS-CP.A.5**
FORWARD 2	Find a conditional probability as the fraction of B's outcomes that…	**HSS-CP.B.6**

SET 58 • 8.SP.A.4

MAIN

Grade 8 main question • Grade 8 • Statistics & Probability • 8.SP.A.4

For each row, what percent of the people use a phone at night?

	Phone at night	No phone at night	Total
Sleeps 8+ hours	12	28	40
Sleeps under 8 hours	36	24	60
Total	48	52	100

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q1/5

Is there evidence of an association between sleep and phone use at night? Explain using the row relative frequencies.

	Phone at night	No phone at night	Total
Sleeps 8+ hours	12	28	40
Sleeps under 8 hours	36	24	60
Total	48	52	100

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q2/5

Which conclusion do the row relative frequencies support?

	Breakfast	No breakfast	Total
Grade 7	30	20	50
Grade 8	45	30	75
Total	75	50	125

A. There is a strong association because 45 is greater than 30.

B. Grade 7 students are more likely to eat breakfast.

C. Grade 8 students are more likely to eat breakfast.

D. There is no association: 60% of each grade eat breakfast.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q3/5

What percent of the people who use a phone at night sleep under 8 hours?

	Phone at night	No phone at night	Total
Sleeps 8+ hours	12	28	40
Sleeps under 8 hours	36	24	60
Total	48	52	100

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q4/5

True or false?

Grade 8 students are more likely to eat breakfast than Grade 7 students, because 45 is greater than 30.

	Breakfast	No breakfast	Total
Grade 7	30	20	50
Grade 8	45	30	75
Total	75	50	125

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q5/5

SET 58 • 8.SP.A.4

BACKWARD 1

Find a percent of a quantity and the percent one number is of another • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

What percent of 40 is 12?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q1/5

Find 60% of 75.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q2/5

36 out of 48 is what percent?

A. 75%

B. 48%

C. 12%

D. 36%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q3/5

True or false?

15 out of 60 is 15%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q4/5

True or false?

20% of 50 is 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q5/5

SET 58 • 8.SP.A.4

BACKWARD 2

Use ratio language to describe two quantities • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1

In a group, 12 people use a phone at night and 28 do not.

Write the ratio of users to non-users in simplest form.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q1/5

A class has 18 students who walk to school and 12 who ride.

For every 3 students who walk, how many ride?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q2/5

In a survey, 30 people eat breakfast and 20 do not.

Which statement describes the ratio?

A. For every 3 who eat breakfast, 2 do not.

B. For every 1 who eats breakfast, 30 do not.

C. For every 3 who eat breakfast, 5 do not.

D. For every 2 who eat breakfast, 3 do not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q3/5

True or false?

The ratio 24 : 16 is equivalent to 3 : 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q4/5

True or false?

The ratio 10 : 15 is equivalent to 3 : 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q5/5

SET 58 • 8.SP.A.4

FORWARD 1

Recognize conditional probability and independence in everyday situations • High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5

Explain the difference between P(sleeps under 8 hours | uses a phone at night) and P(uses a phone at night | sleeps under 8 hours).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q1/5

A fair coin is flipped twice, and the flips are independent.

Is P(second flip is heads | first flip is heads) equal to P(second flip is heads)? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q2/5

Which pair of events is most likely independent?

A. Smoking and getting lung disease

B. Rolling a 6 on a fair number cube and flipping heads on a fair coin

C. Rain today and carrying an umbrella today

D. Studying more and getting a higher grade

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F1-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q3/5

True or false?

If two events are independent, knowing that one happened does not change the probability of the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q4/5

True or false?

P(A | B) is always equal to P(B | A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q5/5

SET 58 • 8.SP.A.4

FORWARD 2

Find a conditional probability as the fraction of B's outcomes that are also in A • High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6

A fair number cube numbered 1 to 6 is rolled.

Given that the roll is even, what is the probability that it is greater than 3?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q1/5

Two fair coins are flipped, and the flips are independent.

Given that at least one coin shows heads, what is the probability that both show heads?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q2/5

Of 100 people, 48 use a phone at night, and 36 of those sleep under 8 hours.

A night phone user is chosen at random. What is P(sleeps under 8 hours | uses a phone at night)?

A. 0.75

B. 0.36

C. 0.48

D. 0.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q3

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q3/5

True or false?

A card is drawn at random from cards numbered 1 to 10.

Given that the number is odd, the probability that it is 5 or less is $\frac{3}{5}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q4/5

True or false?

A fair number cube numbered 1 to 6 is rolled.

Given that the roll is less than 3, the probability that it is 1 is $\frac{1}{6}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q5/5